

Human-AI Codecision Systems

Adaptive decision support under uncertainty, confidence, risk, and contestable authority

Tier 1 - Capstone Research Paper, revised against Synthetic Stage One

Human-AI Codecision Systems Research Program

Stage One revision - September 2026

Evidence status: Synthetic Stage One contains five independent seeds, 180 trials per policy and seed, 3,600 total decisions, and a controlled AI distribution shift in the final quarter. There were zero human participants. Adaptive codecision selectively regulated authority but did not improve synthetic accuracy or declared utility over human-only control. Human trust, satisfaction, and real cognitive state remain unmeasured.

Abstract

The project investigates when an AI system should defer, recommend, assist, or override within a shared decision process. The implemented Synthetic Stage One laboratory operationalizes human-like behavioral telemetry, AI confidence and entropy, risk-sensitive threshold policies, contestability, and authority allocation in a simulated cybersecurity classification task. Across five independent seeds and 3,600 synthetic decisions, adaptive codecision escalated 8.22% of trials and overrode 0.33%, demonstrating selective authority regulation. However, it did not improve synthetic accuracy or the declared utility score over the human-only control. Aggressive automation produced only a small accuracy increase while eliminating autonomy and generating a 21.11% unnecessary-override rate. These results validate the authority-allocation mechanism but reject the current claim that the chosen adaptive thresholds improve outcomes. Human trust and satisfaction were not measured.

1. Research question

Under what observable conditions should an AI system defer, recommend, assist, or override a human decision, and can those authority shifts improve outcomes without excessive loss of human autonomy?

2. Implemented Stage One

- Four policies: human only, static recommendation, adaptive codecision, and aggressive automation.
- Synthetic human responses include decision, confidence, latency, hesitation, revisions, workload, and recent-error history.
- Synthetic AI responses include decision, probability, confidence, entropy, and an explanation tied to the strongest event feature.
- Adaptive authority uses uncertainty, AI confidence, predicted risk, critical-risk thresholds, and contestability.
- Controlled AI distribution shift is introduced during the final quarter of each run.
- Evidence exports preserve per-decision telemetry, per-seed summaries, aggregate summaries, configuration/provenance, and explicit synthetic-study labeling.

3. Frozen evidence

Policy	Accuracy	Escalation	Override	Unnecessa ry	Autonomy	Utility
Human only	0.5589 ± .0606	0	0	0	1.0000	0.6186 ± .0253
Static recommen dation	0.5611 ± .0580	0	0	0	0.7990	0.5834 ± .0246

Adaptive codecision	0.5600 ± .0575	0.0822 ± .0182	0.0033 ± .0050	0.0011 ± .0025	0.7701	0.5787 ± .0268
Aggressive automation	0.5633 ± .0554	1.0000	1.0000	0.2111 ± .0283	0.0000	0.3824 ± .0296

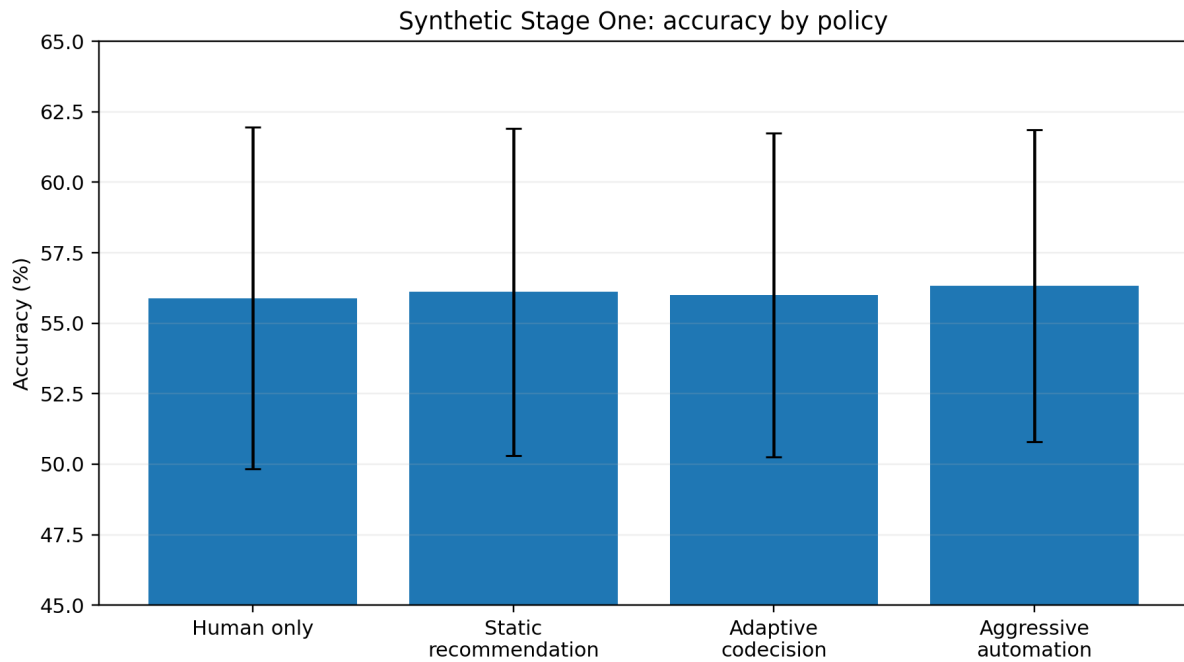


Figure 1. Measured synthetic accuracy $\pm$ sample SD across five independent seeds.

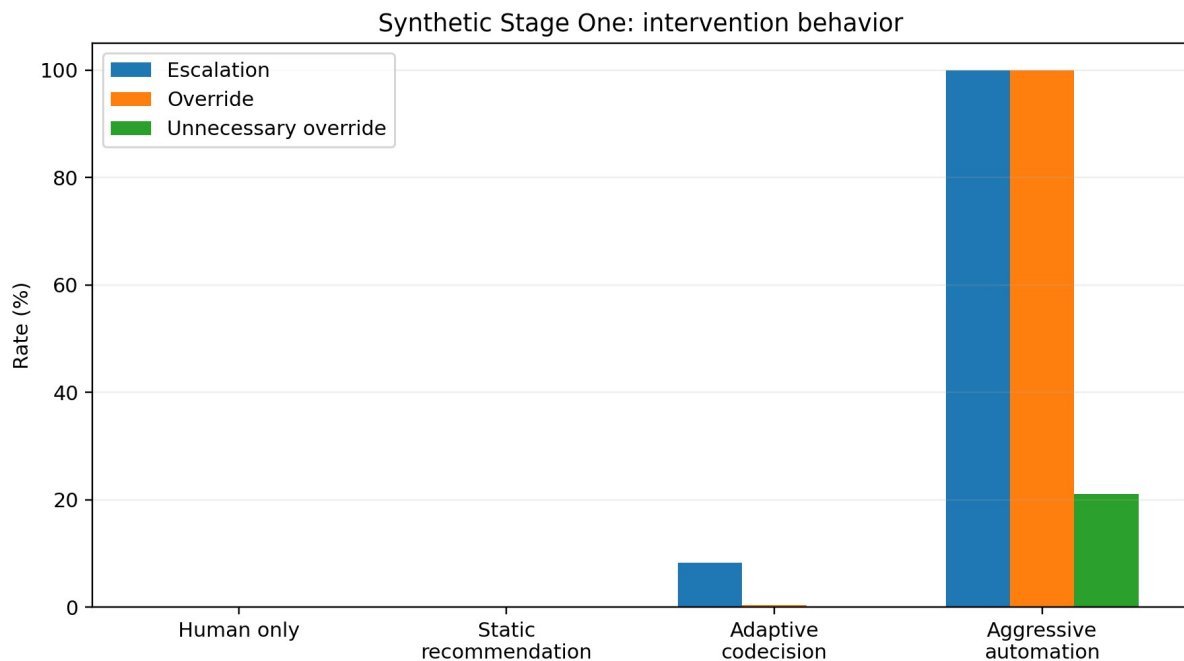


Figure 2. Measured intervention behavior. Adaptive codecision is selective; aggressive automation intervenes on every trial.

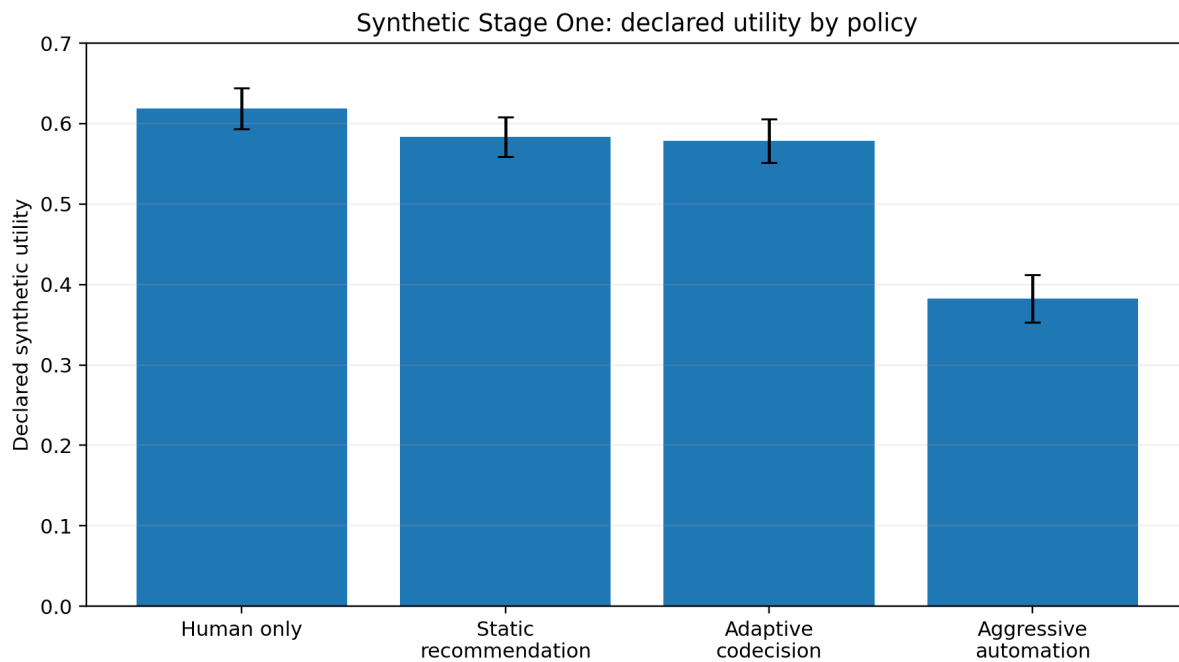


Figure 3. Measured declared synthetic utility $\pm$ sample SD. The current adaptive thresholds do not outperform human-only control.

4. Revised interpretation

The strongest supported Stage One conclusion is mechanistic rather than outcome-superiority: the adaptive policy can selectively redistribute authority. The original hypothesis that adaptive intervention would outperform passive or automated alternatives is not supported by the present synthetic configuration. The next study must separate policy quality from simulated-agent quality through skill sweeps, threshold ablations, and held-out seeds.

5. Claim boundary

Stage One contains no human participants and therefore cannot establish human trust, satisfaction, cognitive workload, behavioral uncertainty validity, reliance, or perceived autonomy. Simulated trust/autonomy variables are model state, not human-subject evidence. The cybersecurity task is a controlled experimental domain, not evidence of production security performance.